

# BestSupport Application Workflow Documentation

Scope: application logic

## 1. Application Purpose

BestSupport is a Django helpdesk application for creating, assigning, approving, discussing, resolving, and closing support work. The main business flows are normal helpdesk tickets, CBS access requests, remote access requests, incident reports, ticket chat, audio-call signaling, technical documents, and portal announcements.

## 2. Main Application Areas

| Area          | Main Files                                                                                                                                                                                                         | Business Responsibility                                                                                                                     |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| Project shell | <code>helpdesk/helpdesk/urls.py</code> , <code>health.py</code>                                                                                                                                                    | Root URL routing, health endpoints, and <code>asgi.py</code> HTTP/WebSocket application entry points.                                       |
| Accounts      | <code>accounts/models.py</code> , <code>forms.py</code> , <code>reset</code> , <code>branches</code> , <code>departments</code> , and <code>active-session</code> <code>views.py</code> , <code>backends.py</code> | User profile data, authentication, signup, password tracking.                                                                               |
| Tickets       | <code>tickets/models.py</code> , <code>forms.py</code> , <code>views.py</code>                                                                                                                                     | Ticket lifecycle, approvals, incident reports, chat, calls, <code>consumers.py</code> queues, documents, notifications, and status changes. |
| Templates     | <code>templates/base.html</code> , <code>templates/tickets/*.html</code> , <code>views</code> , <code>templates/accounts/*.html</code>                                                                             | User-facing pages rendered by the account and ticket                                                                                        |
| Static UI     | <code>static/css/local-ui.css</code> , <code>behavior</code> .                                                                                                                                                     | Shared interface styling and small browser-side <code>static/js/local-ui.js</code>                                                          |

## 3. Request Routing Flow

- A browser request reaches Django.
- `helpdesk/helpdesk/urls.py` routes it to `admin`, `accounts`, `tickets`, `health`, or the root login redirect.
- Account routes are handled through `accounts.urls`; ticket routes are handled through `tickets.urls`.
- The matched view validates login state, permissions, form data, and model state.
- The view reads or updates models, sends notifications or email when needed, and renders a template or redirects.
- For live chat, audio-call signaling, and browser notifications, WebSocket requests are routed through `tickets/routing.py` to consumers in `tickets/consumers.py`.

## 4. Key Models

| Model                    | App Logic Role                                                                                                                                                                |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CustomUser               | User account plus support flags, Central Operation flag, branch, department, position, phone, and signature image.                                                            |
| Branch                   | Named branch records used for user profiles, ticket routing, and document visibility.                                                                                         |
| Department               | Named departments used for ticket routing, support queues, and document visibility.                                                                                           |
| Ticket                   | Main work item. Stores requester, assignee, request type, subject, description, branch, department, priority, status, resolution, closure, chat privacy, and display helpers. |
| RemoteAccessApproval     | Approval record for remote access and CBS access requests. Tracks requester, recommender, second recommender, approver, signatures, status, and decision notes.               |
| TicketMessage            | Text message in a ticket conversation.                                                                                                                                        |
| TicketMessageAttachment  | File metadata linked to a ticket chat message, including browser-viewable image/PDF helpers.                                                                                  |
| TicketChatReadState      | Tracks when each user last saw chat messages for unread indicators.                                                                                                           |
| TicketAssignmentLog      | Assignment and ownership history for reporting and display.                                                                                                                   |
| IncidentReport           | Structured incident report linked to an incident ticket, including report fields, signers, signatures, submission, correction, and commander data.                            |
| IncidentReportSignoff    | Ordered signoff rows for notified incident report reviewers.                                                                                                                  |
| IncidentReportAttachment | Evidence file metadata linked directly to an incident report.                                                                                                                 |
| TechnicalDocument        | Knowledge/document entry with branch, department, support-only, or restricted visibility.                                                                                     |
| PortalFlashAnnouncement  | Login/portal announcement shown to users during a configured schedule.                                                                                                        |
| GroupMailboxEmail        | Shared mailbox mapping used for department-aware ticket email behavior.                                                                                                       |

## 5. Account Workflow

1. A user signs in through `/accounts/login/`.
2. `login_view` authenticates through the configured authentication backend.
3. After login, users are redirected into the ticket/dashboard area.
4. Signup and password reset routes are controlled by authentication feature flags.
5. User department, branch, support role, Central Operation role, and signature image affect permissions throughout ticket, CBS, and incident workflows.

## 6. Normal Ticket Workflow

1. User opens `/tickets/new/`.

2. `create_ticket` builds `TicketForm` for the current user.
3. The form validates request type, department, branch, notify email, assignment email, impact, urgency, CC emails, and attachments.
4. On valid submit, a `Ticket` is created with a generated ticket ID and calculated priority.
5. If an assignment email is supplied, the matching user becomes the assignee. If only notify email is supplied, that address receives notification but does not own the ticket.
6. Department users can see department-routed tickets and may claim unassigned tickets when allowed.
7. Support users work tickets through support queues and the ticket update page.
8. Resolved tickets record resolution note, resolved user, resolved timestamp, and email the requester with a close link.
9. Closed tickets become read-only for chat and cannot be reopened through the normal ticket panel.

## 7. Ticket Status Rules

| Status Area         | Important Rule                                                                                                                                                                |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Assignment          | Unassigned department tickets may be claimed by eligible department users. Created-by users cannot claim their own ticket unless they are also assigned through the workflow. |
| Resolution          | A ticket must be assigned before it can be resolved. Normally, only the assigned agent can mark it resolved.                                                                  |
| Closure             | A ticket can be closed only after it is resolved. Closing is restricted to an admin user or effective assignee.                                                               |
| Resolved/closed     | Resolved and closed tickets cannot be reopened through the normal ticket update reopening panel.                                                                              |
| Incident resolution | Incident tickets require required incident report conditions before resolution is accepted.                                                                                   |
| Approved CBS        | When an approved CBS access request is resolved, document sending, ticket update, resolution and new chat messages are locked.                                                |

## 8. Ticket Detail Page

`ticket_detail` is the central page for a ticket. Its context decides which blocks and actions the user can see.

| Context Area   | What It Controls                                                                             |
|----------------|----------------------------------------------------------------------------------------------|
| Requester info | Shows requester details when the user is allowed to see them.                                |
| Approval data  | Shows remote access or CBS access request details, signatures, status, and reviewer actions. |
| Incident data  | Shows incident report availability, edit/view actions, and download actions.                 |

|              |                                                                                                                                                    |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Chat state   | Loads visible chat messages, unread state, lock message, upload limits, and WebSocket settings.                                                    |
| Action flags | Controls buttons such as update ticket, claim ticket, approve/reject, assign approved CBS access, send approved document, and manage chat privacy. |
| Attachments  | Shows ticket-level image attachment, chat attachments, CBS request attachments, and incident evidence links according to permissions.              |

## 9. Chat and Call Workflow

1. Ticket detail renders the `Ticket Chat` panel if the current user has chat access.
2. `can_access_ticket_chat` decides chat membership.
3. Normal ticket chat membership is based on requester, assignee, support/admin users, department membership, and private-chat settings.
4. CBS access request chat membership is based on requester, assignee, current approval-chain users, users who already acted, Central Operation, IT support, and admins.
5. Remote access approval requests that are not CBS access requests do not use ticket chat.
6. Chat messages are sent over `TicketChatConsumer`, saved as `TicketMessage`, and broadcast to connected participants.
7. Attachments are uploaded by the ticket attachment upload view and then announced in the chat stream.
8. Audio calls use `TicketCallConsumer` only for signaling. Call permission follows the same chat access helper.
9. Closed tickets disable chat and file upload. Approved CBS access requests also disable chat after the linked ticket is resolved.

## 10. CBS Access Request Workflow

1. Head Office requests start at `/tickets/cbs-access-request/`.
2. Branch requests start at `/tickets/cbs-access-request/branch/`.
3. Both routes use the CBS access request view logic with office-specific fields.
4. `CBSAccessRequestForm` validates requester details, access user, user groups, endorsement, recommender, second recommender for branch requests, approver, and attachments.
5. The app creates a ticket and linked `RemoteAccessApproval`.
6. The current reviewer can recommend or reject. Branch requests can require a second recommendation before final approval.
7. Central Operation can adjust the signoff chain before final approval when allowed.
8. When final approval is recorded, the approved CBS document is generated from saved request and approval data.
9. The requester can assign the approved request to a concerned user and email the signed document while the ticket is still open.
10. After final approval and ticket resolution, CBS document sending and ticket status editing are locked.

### CBS Chat Membership

| User Type                                         | Chat Access                                   |
|---------------------------------------------------|-----------------------------------------------|
| Requester                                         | Can participate.                              |
| Assigned user                                     | Can participate.                              |
| Current recommender, second recommender, approver | Can participate while they are in the current |

approval chain.

Users who already recommended, secondrecommended, or approved Keep access as historical actors.

Central Operation

Can participate for CBS access tickets.

IT support/admin

Can participate.

Removed reviewer who never acted

Does not keep chat access after being removed from the chain.

## 11. Remote Access Request Workflow

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1. User opens `/tickets/remote-access-request/`.
2. `RemoteAccessRequestForm` validates subject, details, recommender, and approver.
3. The app creates a ticket and a linked `RemoteAccessApproval`.
4. The approval record tracks pending recommendation, pending approval, approved, or rejected state.
5. Reviewer decisions are handled by `remote_access_approval_update`.
6. Remote access approval requests do not use the normal ticket chat flow.

## 12. Incident Report Workflow

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1. An incident ticket can have a linked `IncidentReport`.
2. `ticket_incident_report` renders and processes the report form.
3. Support users, the requester, and the incident commander can manage the report when allowed.
4. The report tracks incident information, containment, recovery, lessons learned, action plan, notification data, and evidence notes.
5. Evidence files are listed with view/download actions when the current user can access the incident report.
6. Registered user and notified signoff users sign in sequence according to the saved signoff rows.
7. Incident reports can be saved as draft, submitted, sent to reviewers/approvers, corrected, or locked depending on state.
8. Incident ticket resolution is blocked until required report and signature conditions are complete.

### Incident Report Access

| User Type                     | Access                                                |
|-------------------------------|-------------------------------------------------------|
| Ticket participant            | Can view the incident report.                         |
| Incident commander            | Can view and manage where allowed.                    |
| Registered user               | Can view/sign according to role.                      |
| Notified user or signoff user | Can view/sign according to sequence.                  |
| Support/admin                 | Can view and manage according to support permissions. |

## 13. Technical Documents Workflow

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1. Users open `/tickets/docs/` to see documents visible to them.

2. Support users can upload documents through `tech_docs_upload`.
3. Visibility can be public, branch-based, department-based, restricted to selected users, or supportonly.
4. View and download actions check document visibility before serving content.
5. Supported spreadsheet documents can be previewed in a browser-friendly table view.

## 14. Portal Flash Announcements

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1. Support/admin users upload announcements through the portal flash upload page.
2. Each announcement has title, message, image, visibility, and schedule fields.
3. Active announcements are shown to users after login according to schedule and visibility rules.

## 15. Support Queues

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| Queue                                    | Purpose                                                                                                               |
|------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| <code>support_dashboard</code>           | High-level support dashboard with counts, workload, and recent tickets.                                               |
| <code>support_queue</code>               | Main ticket queue with filters for ticket ID, status, priority, requester, assignee, branch, department, and dates.   |
| <code>support_department_tickets</code>  | Department-focused view split into unassigned, assigned, and solved groups.                                           |
| <code>support_cbs_access_requests</code> | CBS-specific request queue for support/Central Operation users.                                                       |
| <code>support_incident_tickets</code>    | Incident-specific support queue. <code>support_users</code> Currently logged-in users and related support visibility. |

## 16. Important URL Groups

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| URL Pattern                                                                     | Business Use                             |
|---------------------------------------------------------------------------------|------------------------------------------|
| <code>/tickets/</code>                                                          | My Tickets list.                         |
| <code>/tickets/new/</code>                                                      | Create normal ticket.                    |
| <code>/tickets/&lt;ticket_id&gt;/</code>                                        | Ticket detail page.                      |
| <code>/tickets/support/&lt;ticket_id&gt;/update/</code>                         | Ticket assignment/status update.         |
| <code>/tickets/&lt;ticket_id&gt;/claim/</code>                                  | Claim eligible department ticket.        |
| <code>/tickets/&lt;ticket_id&gt;/attachments/upload/</code>                     | Add file to ticket chat.                 |
| <code>/tickets/&lt;ticket_id&gt;/attachments/&lt;attachment_id&gt;/view/</code> | View supported ticket attachment inline. |

|                                                                                     |                                                      |
|-------------------------------------------------------------------------------------|------------------------------------------------------|
| <code>/tickets/&lt;ticket_id&gt;/attachments/&lt;attachment_id&gt;/download/</code> | Download ticket attachment.                          |
| <code>/tickets/cbs-access-request/</code>                                           | Create Head Office CBS access request.               |
| <code>/tickets/cbs-access-request/branch/</code>                                    | Create Branch CBS access request.                    |
| <code>/tickets/&lt;ticket_id&gt;/remote-access-approval/</code>                     | Recommend, approve, or reject remote/CBS request.    |
| <code>/tickets/&lt;ticket_id&gt;/cbs-access-request/senddocument/</code>            | Email approved CBS document while allowed.           |
| <code>/tickets/&lt;ticket_id&gt;/incident-report/</code>                            | Create, view, edit, submit, or sign incident report. |
| <code>/tickets/docs/</code>                                                         | Technical document list.                             |

## 17. Developer Takeover Map

A new developer should treat `tickets/views.py` as the main business-orchestration file and `tickets/models.py` as the source of workflow state and permission helper behavior. Templates decide what the user sees, but important permissions must also be enforced in views and consumers.

| Need To Change                             | Start Here                                                                                                                                                   | Then Check                                                                                                          |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| Ticket creation fields or validation       | <code>tickets/forms.py</code> → <code>TicketForm</code>                                                                                                      | <code>tickets/views.py</code> → <code>create_ticket</code> , then <code>templates/tickets/create_ticket.html</code> |
| Ticket detail actions                      | <code>tickets/views.py</code> → or visibility <code>_build_ticket_detail_context</code>                                                                      | <code>templates/tickets/ticket_detail.html</code>                                                                   |
| Support status                             | <code>tickets/views.py</code> → update rules <code>_ticket_update_protected</code>                                                                           | <code>tickets/forms.py</code> → <code>TicketUpdateForm</code> , <code>TicketAssigneeUpdateForm</code>               |
| Need To Change                             | Start Here                                                                                                                                                   | Then Check                                                                                                          |
| Chat access or                             | <code>tickets/models.py</code> → notification <code>can_access_ticket_chat</code> , <code>participants</code> , <code>get_ticket_chat_access_user_ids</code> | <code>tickets/consumers.py</code> , <code>tickets/notifications.py</code> , <code>tickets/chat_rules.py</code>      |
| CBS access request fields                  | <code>tickets/forms.py</code> → <code>CBSAccessRequestForm</code>                                                                                            | <code>tickets/views.py</code> CBS helper functions and <code>templates/tickets/cbs_access_request.html</code>       |
| CBS approval-chain behavior                | <code>tickets/views.py</code> → <code>remote_access_approval_update</code> , <code>cbs_access_signoff_chain_update</code>                                    | <code>RemoteAccessApproval</code> in <code>tickets/models.py</code>                                                 |
| Incident report fields or signoff behavior | <code>tickets/forms.py</code> → <code>IncidentReportForm</code>                                                                                              | <code>tickets/views.py</code> incident helper functions and <code>templates/tickets/incident_report.html</code>     |
| Document visibility rules                  | <code>tickets/views.py</code> → <code>_can_view_tech_doc</code> , <code>_tech_doc_visibility_q</code>                                                        | <code>TechnicalDocument</code> in <code>tickets/models.py</code>                                                    |

|                   |                                                             |                                                                   |
|-------------------|-------------------------------------------------------------|-------------------------------------------------------------------|
| User role/profile | <code>accounts/models.py</code> → <code>CustomUser</code> , | <code>accounts/views.py</code> , <code>accounts/admin.py</code> , |
| behavior          | <code>Department</code> , <code>Branch</code>               | and relevant ticket permission helpers                            |

## 18. Permission Helpers and Access Rules

The project relies on small helper functions instead of scattering permission logic everywhere. When changing access behavior, update the helper and then confirm every direct URL path still enforces it.

| Helper                                            | Location                                        | Used For                                                                                                                               |
|---------------------------------------------------|-------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| <code>_is_support_user</code>                     | <code>tickets/</code><br><code>views.py</code>  | Identifies staff, superusers, and IT support users for support queues/actions.                                                         |
| <code>_is_central_operation_user</code>           | <code>tickets/</code><br><code>views.py</code>  | Identifies Central Operation users for CBS-specific visibility and actions.                                                            |
| <code>_is_ticket_participant</code>               | <code>tickets/</code><br><code>views.py</code>  | Controls ticket detail and attachment access for requester, assignee, support, department members, CBS reviewers, and incident actors. |
| <code>can_access_ticket_chat</code>               | <code>tickets/</code><br><code>models.py</code> | Controls chat, call signaling, chat message visibility, upload, delete, seen state, and WebSocket connection.                          |
| <code>get_ticket_chat_access_user_ids</code>      | <code>tickets/</code><br><code>models.py</code> | Builds chat and call notification target users.                                                                                        |
| <code>can_manage_ticket_chat_privacy</code>       | <code>tickets/</code><br><code>models.py</code> | Controls whether a ticket's private-chat setting can be changed. Approval tickets do not expose this                                   |
| Helper                                            | Location                                        | Used For                                                                                                                               |
| <code>_can_access_incident_report</code>          | <code>tickets/</code><br><code>views.py</code>  | Controls incident report pages, evidence attachments, signatures, and report downloads.                                                |
| <code>_can_manage_incident_report</code>          | <code>tickets/</code><br><code>views.py</code>  | Controls who can create/update incident report content.                                                                                |
| <code>_can_decide_remote_access_approval</code>   | <code>tickets/</code><br><code>views.py</code>  | Controls recommend/approve/reject buttons and POST behavior for remote/CBS approvals.                                                  |
| <code>_can_manage_cbs_access_signoff_chain</code> | <code>tickets/</code><br><code>views.py</code>  | Controls Central Operation signoff-chain changes before final CBS approval.                                                            |

### Permission Change Rule

1. Never rely only on hiding a button in a template.
2. Enforce the same rule in the view handling the direct URL or POST.
3. If WebSockets are involved, enforce the rule in the consumer access check too.
4. If files are involved, enforce the rule on both view and download URLs.
5. Add tests for both successful access and blocked access.

## 19. State Machines and Business Status



## Ticket Status

| Status                              | Meaning                                                           | Important Behavior                                                                                                     |
|-------------------------------------|-------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| <code>new</code>                    | Ticket has been created and is waiting for action.                | Visible in queues according to owner/department/role.                                                                  |
| <code>acknowledged</code>           | Someone has seen/accepted the work but it is not yet in progress. | Still active.                                                                                                          |
| <code>in_progress</code>            | Work is actively being handled.                                   | Active support workflow.                                                                                               |
| <code>waiting_on_user</code>        | Support is waiting for requester/user input.                      | Active but blocked externally.                                                                                         |
| <code>waiting_on_third_party</code> | Support is waiting for outside/vendor/team input.                 | Active but blocked externally.                                                                                         |
| <code>resolved</code>               | Support considers the issue fixed.                                | Requester receives email with close link. CBS approved requests become locked for document sending and status editing. |
| <code>closed</code>                 | Ticket is complete.                                               | Chat and uploads are disabled. Normal panel cannot reopen.                                                             |
| <code>cancelled_duplicate</code>    | Ticket was cancelled as duplicate/not needed.                     | Treated as a solved state in queues.                                                                                   |

## Remote/CBS Approval Status

| Status                              | Meaning                                                            | Typical Next Step                                                |
|-------------------------------------|--------------------------------------------------------------------|------------------------------------------------------------------|
| <code>pending_recommendation</code> | Request is waiting for first or second recommendation.             | Current recommender approves or rejects.                         |
| <code>pending_approval</code>       | Required recommendation stage is done; final approver must decide. | Approver approves or rejects.                                    |
| <code>approved</code>               | Final approval is complete.                                        | CBS request can be assigned/processed while ticket remains open. |
| <code>rejected</code>               | Reviewer rejected the request.                                     | Requester can correct/resubmit if workflow allows.               |

## Incident Report State

| State                  | How It Is Represented                                              | Effect                                                 |
|------------------------|--------------------------------------------------------------------|--------------------------------------------------------|
| Draft                  | exists but is not submitted. evidence.                             | Allowed managers can edit report fields and Report     |
| Waiting for signatures | Required signature fields or signoff rows signatures are complete. | Submit/resolution may be blocked until are incomplete. |

|                      |                                                                  |                                                         |
|----------------------|------------------------------------------------------------------|---------------------------------------------------------|
| Submitted            | <code>submitted_at</code> and <code>submitted_by</code> are set. | Report is treated as formally sent/submitted.           |
| Correction requested | fields are set. on role and lock rules.                          | Report can be returned for changes depending Correction |
| Finalized/locked     | Workflow lock helpers determine                                  | Editing is blocked through normal views. finality.      |

## 20. Detailed Workflow: Normal Ticket

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1. User opens the create ticket page.
2. The form preloads user context such as available departments, branches, and support routing options.
3. User selects request type, department/branch, impact, urgency, notify email, assignment email, CC emails, description, and optional attachments.
4. The form validates email behavior. Assignment email must map to a real user and must not be a group mailbox. Notify email can be individual or shared mailbox according to rules.
5. Priority is calculated from impact and urgency.
6. The ticket is created with requester set to the current user.
7. Assignment is applied if requested. Otherwise, department routing and support queues determine visibility.
8. Notification email is sent to configured recipients.
9. Requester lands on the ticket detail or list page and can continue in chat.
10. Support users pick up work from support queues, claim if eligible, update status, and resolve/close according to rules.

## 21. Detailed Workflow: CBS Access Request

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1. Requester opens either Head Office or Branch CBS request form.
2. The form captures user details, requested CBS user groups, request type, amendment reason where applicable, access user, signoff users, and supporting attachments.
3. On submit, the app creates a ticket with request type `cbs_access` or `cbs_access_branch`.
4. The app creates a linked `RemoteAccessApproval` with recommender, optional second recommender, and approver.
5. The request detail appears inside the ticket detail page using saved ticket and approval data.
6. The current reviewer receives the request and can recommend/reject.
7. For branch requests, a second recommendation can be required before final approval.
8. The final approver approves or rejects.
9. Approved request displays signed approval information and allows allowed users to send/assign the approved document while the ticket remains active.
10. CBS request chat is available to current chain users, historical signers, requester, assignee, Central Operation, support, and admins.
11. If Central Operation changes the chain before a reviewer acts, the old reviewer loses chat access. If the old reviewer already acted, they keep access as a historical signer.
12. Once approved and the linked ticket is resolved, approved document sending, ticket update, and new chat messages are locked.

## 22. Detailed Workflow: Incident Report

1. Incident ticket is created or selected.
2. Allowed user opens the incident report page from ticket detail.
3. The report form captures incident timeline, service impact, branch impact, containment, recovery, root cause, lessons learned, recommendations, notification details, and action plan.
4. Evidence files can be uploaded and later viewed/downloaded by users who can access the incident report.
5. Registered user signature is copied from the user's profile signature when available.
6. Notified signoff rows define ordered reviewers/signers.
7. Signoff users can sign or reject according to sequence and role.
8. Submit sends the report to configured recipients when required fields/signatures are complete.
9. Incident ticket resolution checks report completion rules first.
10. After finalization, normal editing is blocked.

## 23. File View and Download Logic

This section documents only application permission behavior for files. It does not describe file backend implementation or infrastructure.

| File Type                    | Where User Sees It                                                                                   | Access Rule                                                                                             |
|------------------------------|------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| Ticket image                 | Ticket detail                                                                                        | Visible to ticket participants.                                                                         |
|                              | attachment block                                                                                     |                                                                                                         |
| Ticket message attachment    |                                                                                                      | Visible to chat participants. CBS access request participants can                                       |
|                              | Ticket chat and CBS view request attachments even if they are not normal chat attachment list users. |                                                                                                         |
| Incident evidence attachment | Incident report page                                                                                 | Visible to users who can access the incident report.                                                    |
|                              |                                                                                                      |                                                                                                         |
| Technical document           | Technical documents                                                                                  | Visible according to document visibility: public, branch, page department, restricted, or support-only. |

## 24. Email and Notification Logic

| Trigger             | What Happens                                                                                                                   |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Ticket created      | Requester/support recipients receive new ticket notification based on assignment and notify-email rules.                       |
| Ticket assigned     | Assigned user is notified and may receive assignment email.                                                                    |
| Ticket resolved     | Requester receives resolution email with close link, resolution details, optional CC, and optional resolved-email attachments. |
| Ticket chat message | Eligible chat participants except the sender receive browser notification payloads.                                            |
| Ticket audio call   | Eligible chat/call participants receive incoming call notification payloads.                                                   |

Remote/CBS approval stage changes      Current reviewer or approver receives email/notification where configured.

Approved CBS request assigned      Concerned user receives assignment email with signed document attached.

Incident report submit/signoff      Relevant reviewers, approvers, signers, and CC recipients receive report-related email/notification.

## 25. Template Ownership

| Template                                               | Purpose                                                                                                    | Key Context Variables                                                                                                                                                                                        |
|--------------------------------------------------------|------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>templates/base.html</code>                       | Global layout, sidebar,                                                                                    | User, auth flags, active navigation, flash                                                                                                                                                                   |
| Template                                               | Purpose                                                                                                    | Key Context Variables                                                                                                                                                                                        |
|                                                        | header, notifications, and shared page frame.                                                              | announcements.                                                                                                                                                                                               |
| <code>templates/tickets/ticket_list.html</code>        | User ticket list and filters.                                                                              | <code>tickets</code> , selected filters, unread flags, claim/update flags.                                                                                                                                   |
| <code>templates/tickets/create_ticket.html</code>      | Normal ticket create form.                                                                                 | <code>form</code> , field choices, validation errors.                                                                                                                                                        |
| <code>templates/tickets/ticket_detail.html</code>      | Main ticket detail, CBS details, incident link, chat, attachments, assignment history, and actions.        | <code>ticket</code> , <code>remote_access_approval</code> , <code>can_update_ticket</code> , <code>can_view_chat</code> , <code>chat_messages</code> , <code>ticket_attachments</code> , CBS/incident flags. |
| <code>templates/tickets/ticket_update.html</code>      | Support/assignee update form.                                                                              | <code>ticket</code> , <code>form</code> .                                                                                                                                                                    |
| <code>templates/tickets/cbs_access_request.html</code> | Head Office and Branch CBS <code>form</code> , office type flags, signoff fields. access request forms.    |                                                                                                                                                                                                              |
| <code>templates/tickets/incident_report.html</code>    | Incident report form, evidence list, signoff <code>incident_attachments</code> , controls, submit/download | <code>form</code> , <code>incident_report</code> , signoff flags, lock flags. actions.                                                                                                                       |
| <code>templates/tickets/support_queue.html</code>      | Main support ticket queue.                                                                                 | Filtered tickets and selected filter values.                                                                                                                                                                 |
| <code>templates/docs/tech_docs.html</code>             | Technical document listing. <code>documents</code> , visibility and action flags.                          |                                                                                                                                                                                                              |

## 26. Testing Guidance for New Developers

The most important tests live in `tickets/tests.py` and `accounts/tests.py`. When changing workflow behavior, add focused tests near the related test class.

| Change Type       | Test What                                                                                                                       |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------|
| Permission change | Allowed user succeeds, blocked user fails, direct URL/POST path enforces the rule, and template does not show forbidden action. |

|                   |                                                                                                                                                                              |                                         |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
| Status transition | Valid transition succeeds, invalid transition is blocked, saved fields are correct, and change message is shown.                                                             | user-facing                             |
| CBS approval      | Each approval stage, final approval/rejection, signoff-chain changes, requester change                                                                                       | correction, and resolved-lock behavior. |
| Incident report   | Draft save, submit, signatures, signoff order, evidence access, resolution blockers, and change behavior.                                                                    | finalized lock                          |
| Change Type       | Test What<br>HTTP detail visibility, WebSocket access helper impact, attachment upload, delete, Chat change read-state, notification targets, and closed/read-only behavior. |                                         |
| Template action   | Button/link appears for allowed user and is absent for disallowed user; direct backend change be tested.                                                                     | route must also                         |

## 27. Safe Development Checklist

1. Identify the URL and view first.
2. Find the form and model involved.
3. Find existing permission helpers before adding new checks.
4. Update both UI visibility and server-side enforcement.
5. For chat/call changes, confirm WebSocket consumer permissions still match HTTP permissions.
6. For attachments/documents, confirm view and download URLs use the same intended access rule.
7. For approval or incident changes, test both the current actor and a user who should not be allowed.
8. Keep status-locking rules explicit; avoid changing resolved/closed behavior accidentally.
9. Add or update focused tests before broader refactors.
10. Do not mix unrelated UI, workflow, and data-model changes in one patch unless the feature truly needs all three.

## 28. How to Understand or Extend a Feature

1. Start with `helpdesk/helpdesk/urls.py` to find the application route.
2. Open the matching app `urls.py` to identify the view function.
3. Read the view in `views.py` for permission checks, model queries, form validation, state changes, notifications, email, and template context.
4. Read the form in `forms.py` for user-facing fields and validation rules.
5. Read the model in `models.py` for stored fields, helper properties, status choices, and relationships.
6. Open the rendered template to understand which actions are visible to each user.
7. For real-time behavior, read `asgi.py`, `tickets/routing.py`, and `tickets/consumers.py`.
8. For permission-sensitive changes, add tests around both the UI visibility and the direct URL/POST path.